

Vegetation classification in remote sensing

A practical route from labelled data to Random Forest, deep learning and foundation models

From a supervised classification workflow to current GeoAI

with emphasis on vegetation and environmental applications

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Training data

Random Forest

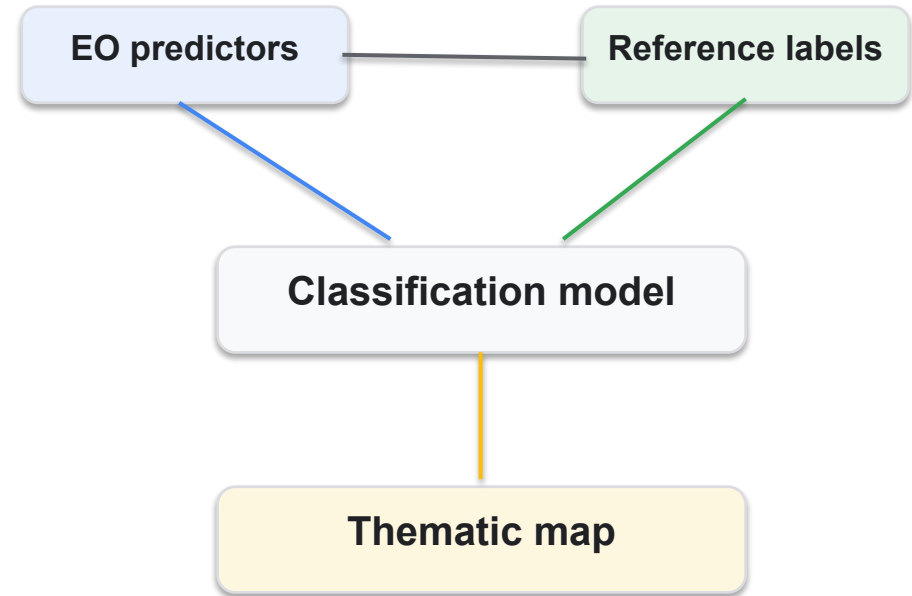
Deep learning

Foundation models

What is a remote-sensing classification?

The basic task

- Assign a discrete class to each pixel, object or image patch.
- Classes should be ecologically meaningful and separable with the available data.
- The result is a thematic map, not simply a model score.



In vegetation studies, the class may represent cover, community, habitat, disturbance state or a broader land-surface type.

Predictors in, class labels out

Predictors (X)

- Spectral bands: e.g. B2, B3, B4, B8.
- Derived variables: e.g. NDVI or red-edge indices.
- Optional additions: texture, topography, SAR or temporal metrics.
- Each predictor should add information relevant to the class problem.

Labels (y)

- Field observations, mapped polygons, expert interpretation or a trusted reference layer.
- Labels define what the model is actually trained to reproduce.
- Class definitions must be internally consistent.
- Poor labels limit every classifier, including deep learning.

Field context: vegetation and landscape setting in Adventdalen

Why ecological and terrain context matters for vegetation classification



Valley-scale landscape context



Local vegetation and wet-ground context



Strong topographic gradients from valley floor to mountain slopes



Fine-scale moisture variation from wet depressions and channels to drier ground



Sparse Arctic tundra vegetation mixed with bare or sparsely vegetated surfaces



Snow persistence, drainage and terrain position influence vegetation patterns

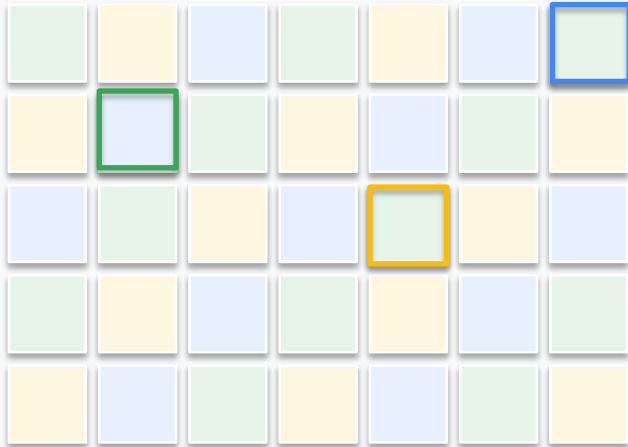


Field context helps interpret both spectral signals and classification errors

How do we construct training data?

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1 Label representative examples



Training samples should cover the range of conditions expected in the mapping area.

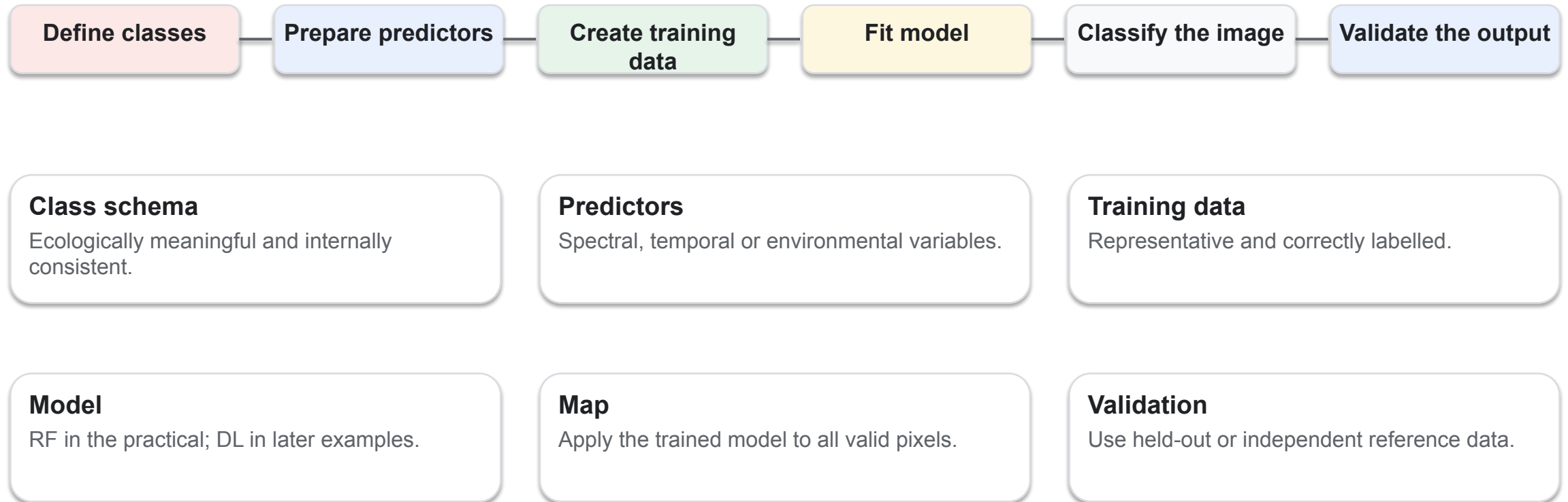
2 Extract predictor values

3 Check class balance

4 Separate train and test data

Important risk: spatial leakage. Nearby training and test pixels can make validation look unrealistically good.

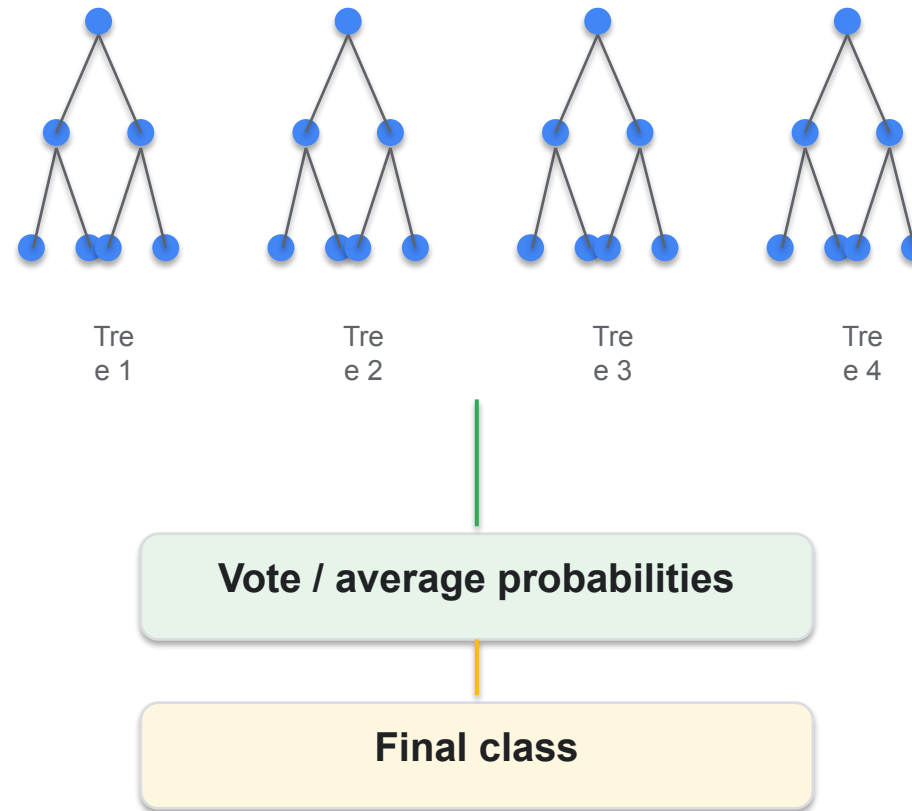
A simple supervised classification workflow



Random Forest — the basic idea

What it learns

- A decision tree repeatedly splits the training samples using predictor thresholds.
- Random Forest fits many different trees.
- Each tree sees a random subset of samples and predictors.
- Predictions are combined across the ensemble.



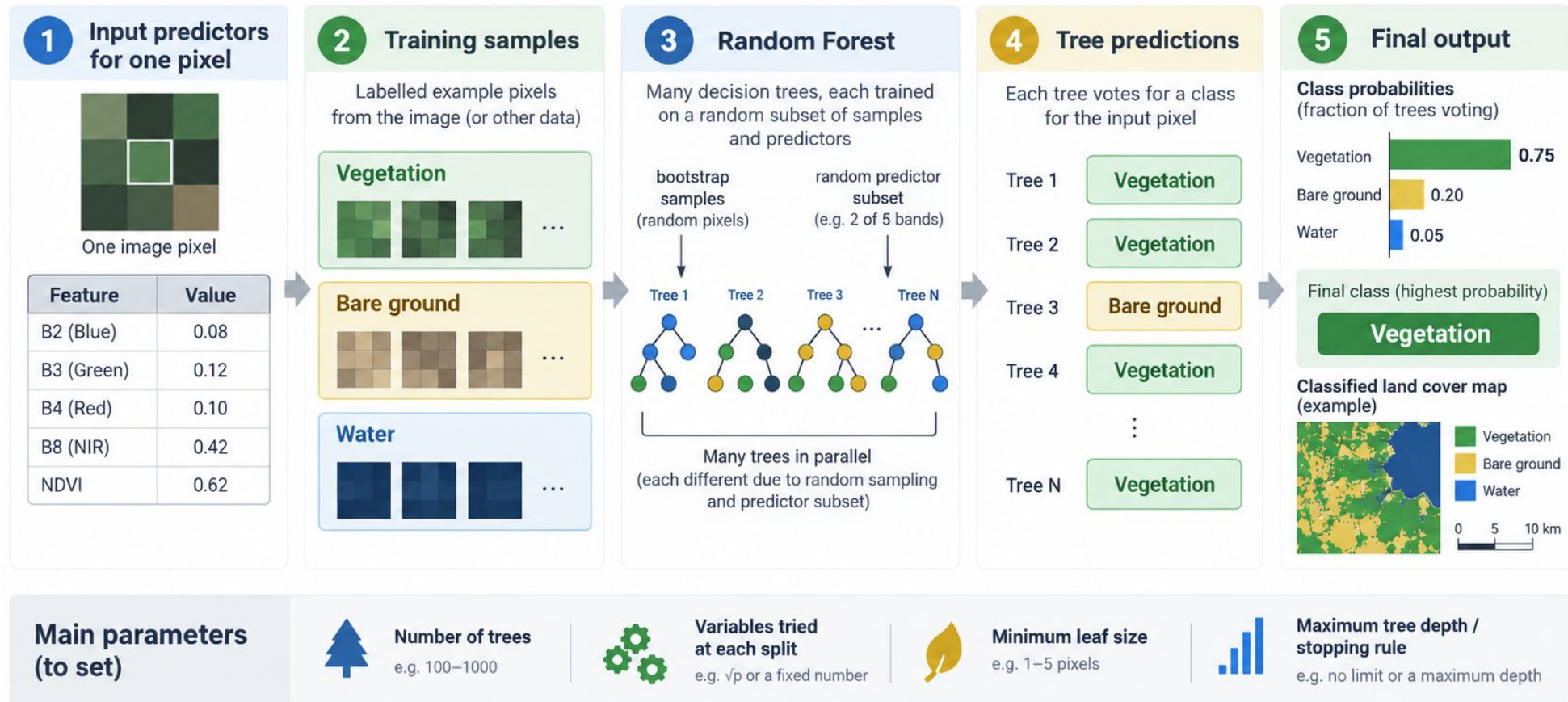
RF uses one feature vector per sample. It does not learn image neighbourhoods unless spatial predictors are supplied.

Random Forest classification — the practical workflow

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Random Forest classification in remote sensing

From pixel values to a land cover map using an ensemble of decision trees



RF learns decision rules from labelled examples; it does not use image neighbourhoods unless spatial features are supplied.

For the practical: B2, B3, B4, B8 and NDVI → labelled pixels → 100-tree Random Forest → three broad classes.

Random Forest in today's practical — inputs, parameters and outputs

Inputs

- One Sentinel-2 summer acquisition.
- Predictors: B2, B3, B4, B8 and NDVI.
- Broad labels: vegetation, bare / sparse ground, water.

Main parameters

- Number of trees: 100.
- Random seed for reproducibility.
- Default tree settings are sufficient for this introductory example.

Outputs

- A classified map for the Adventdalen AOI.
- Class probabilities or votes.
- Variable importance as a descriptive diagnostic.

How we read the result

- Look at the map itself.
- Check the confusion matrix.
- Report overall accuracy, precision, recall, F1 and IoU.
- Inspect which classes are confused.

Validation — how do we assess the classification?

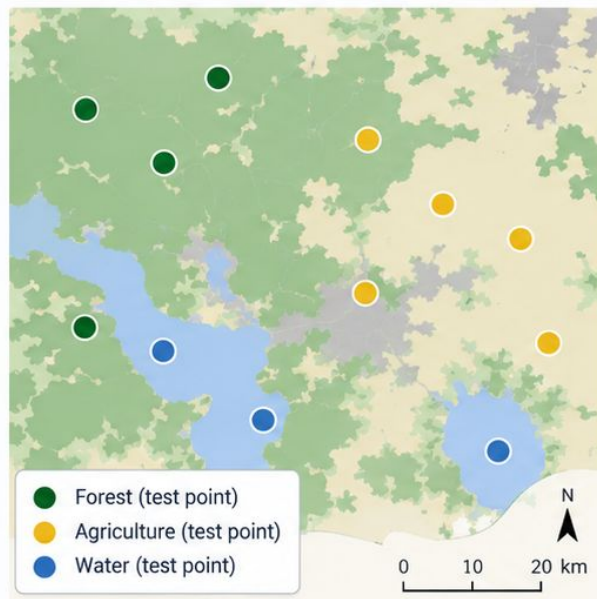
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Validating a Remote-Sensing Classification

Use independent test data to assess how well the model predicts land cover.

1 Reference test data

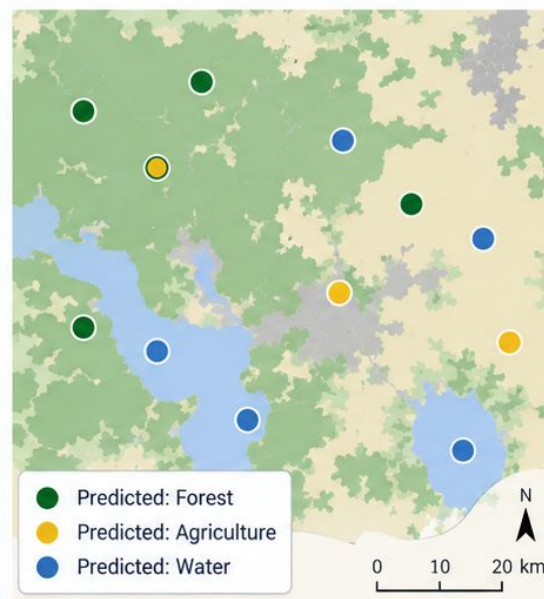
Labelled test points, spatially separate from training



Test points are spatially separate from training data (e.g. different regions or spatial blocks).

2 Model prediction

Predicted labels for the same test points



Same test points as in (1), coloured by the model's predicted class.

3 Compare

Confusion matrix (example with 3 classes)

		Predicted class		
		Forest	Agriculture	Water
Reference class	Forest	45	4	1
	Agriculture	6	38	6
	Water	2	5	42

Overall accuracy = all correct / all test samples

Precision = how reliable a predicted class is (of all pixels predicted as class X, what fraction are correct?)

Recall = how much of a reference class is retrieved (of all reference pixels of class X, what fraction are correctly found?)

F1 score = balance of precision and recall ($F1 = 2 \cdot \text{precision} \cdot \text{recall} / (\text{precision} + \text{recall})$)

IoU = overlap between prediction and reference (IoU = intersection / union)



In Earth observation, random neighbouring pixels can give over-optimistic results.

Prefer spatially separated test areas or spatial blocks (e.g. different regions, tiles or time periods) to obtain an independent and realistic assessment.

Validation compares held-out or independent reference data with the model prediction. In EO, spatial separation matters.

What should we watch carefully?

Data and labels

- Clouds, shadows or seasonal mismatch.
- Ambiguous or inconsistent labels.
- Class imbalance.

Model and validation

- Spatial leakage between train and test data.
- Over-reliance on overall accuracy alone.
- Confusion near boundaries or mixed pixels.

Interpretation

- A good map is not only accurate; it should also make ecological sense.
- Always relate map patterns to the data, the sampling design and known sources of uncertainty.

Machine learning and deep learning — what changes?

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Machine learning (e.g. Random Forest)

Satellite image



0 1 2 km

Label: built-up

Label: forest

Label: water

... more labelled pixels/objects

Feature table (one row = one sample)

ID	B2	B3	B4	B8	NDVI	slope	label
1	0.12	0.15	0.20	0.36	0.42	5	forest
2	0.08	0.10	0.12	0.18	0.33	2	water
3	0.21	0.28	0.31	0.40	0.19	8	built-up
4	0.11	0.14	0.18	0.32	0.37	6	forest
5	0.07	0.09	0.11	0.16	0.28	3	water
...	...	...	...	...	...	...	...

Random Forest



Class prediction

Input unit: pixel or object + predefined predictors

Deep learning

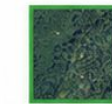
Satellite image divided into patches



Patch labels (examples)



Label: built-up



Label: forest



Label: water

... more labelled patches or masks

CNN / Transformer



Patch/Pixel prediction
(e.g. land cover map)

Input unit: image patch or sequence; features are learned by the model

Both need representative labels and independent validation.



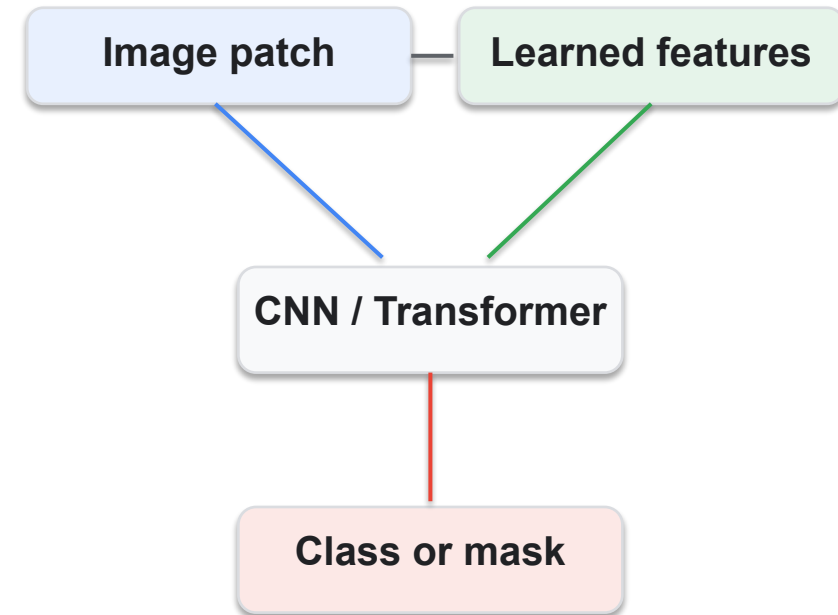
Avoid spatial leakage: nearby training and test samples are not independent.

Conventional machine learning uses analyst-defined predictors; deep learning learns feature representations from image patches or tensors.

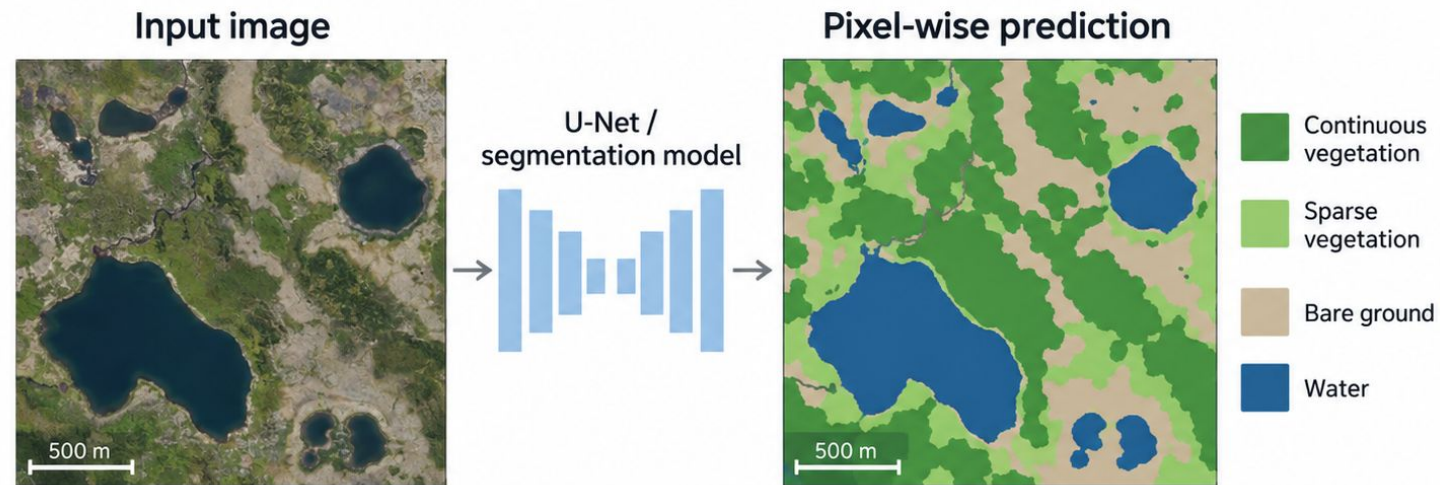
Deep learning — the core idea

What changes compared with RF?

- The model learns feature representations directly from the imagery.
- Spatial context is used explicitly, not only through handcrafted features.
- The input is often an image patch, tile or dense tensor rather than one row in a feature table.



Benefit: richer spatial representation. Cost: more data, more computing and usually less direct interpretability.



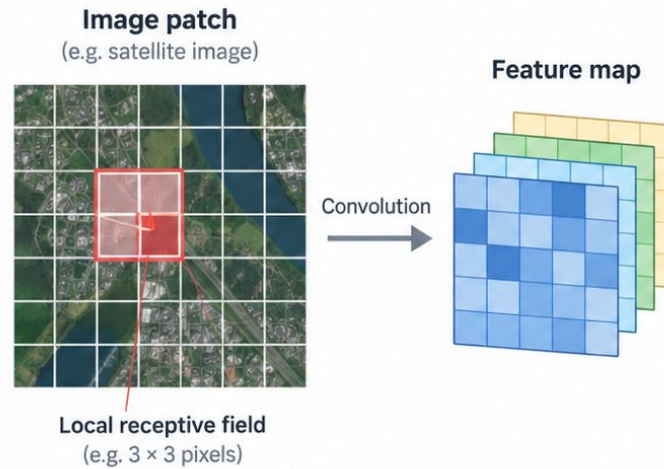
Semantic segmentation predicts a class for every pixel.

Semantic segmentation predicts a class for every pixel and is therefore especially relevant for thematic mapping.

CNNs and Transformers — two common deep-learning families

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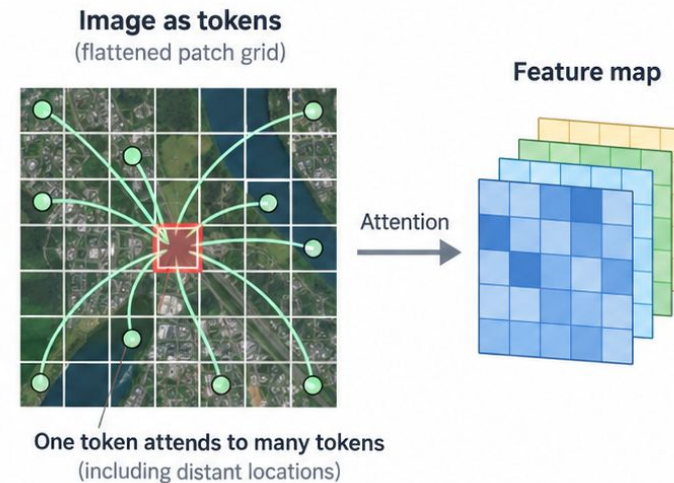
Convolutional neural network



Local spatial context

Each output uses information from a small neighbourhood of nearby pixels.

Transformer / attention



Long-range context

Each output can use information from any location in the image.

Deep models differ in how they represent spatial relationships.

CNNs build context through local receptive fields; attention-based models can capture wider relationships more directly.

Machine learning and deep learning — a practical comparison

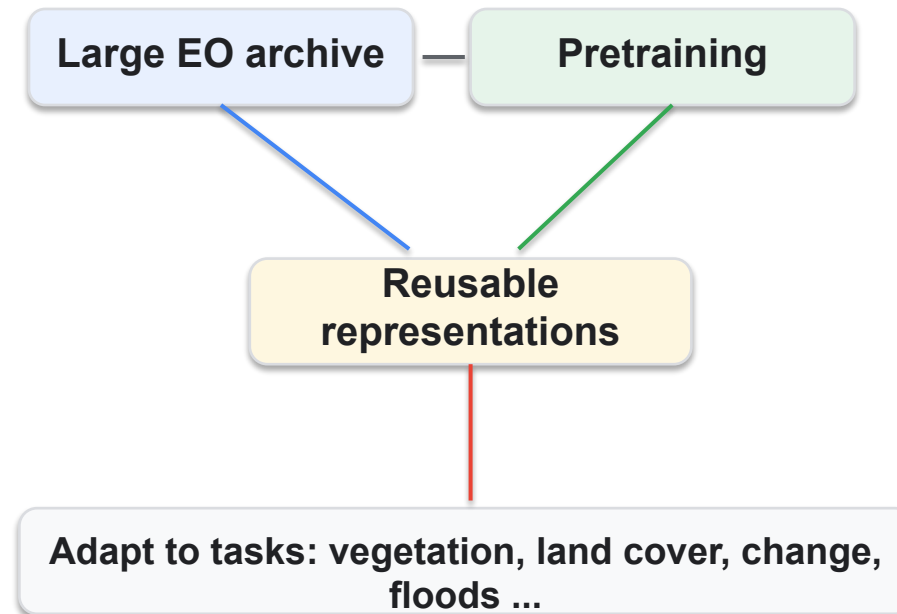
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Aspect	Machine learning (e.g. RF)	Deep learning
Input unit	Feature vector for one pixel or object	Patch, tile, sequence or dense tensor
Feature design	Mainly analyst-defined	Mainly learned by the model
Spatial context	Indirect unless supplied	Explicit and learned
Data demand	Often moderate	Often higher
Computing demand	Usually modest	Usually higher
Typical strength	Transparent, robust baseline	Richer spatial representation

What is a foundation model in Earth observation?

Working definition

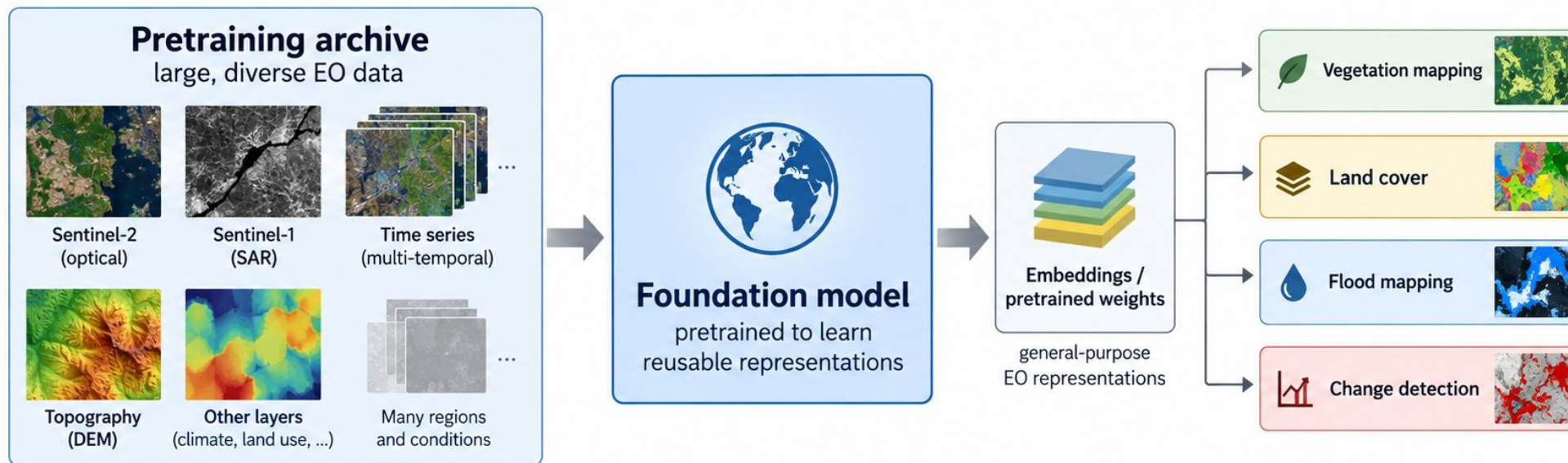
- A large model pretrained on a broad EO archive.
- It is designed to learn reusable representations rather than one single task.
- The pretrained model can then be adapted to specific downstream tasks.



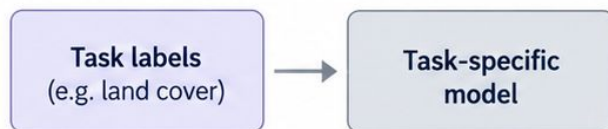
Why develop them? To reuse information learned from large EO archives when labelled task data are limited.

Foundation models — the broader EO workflow

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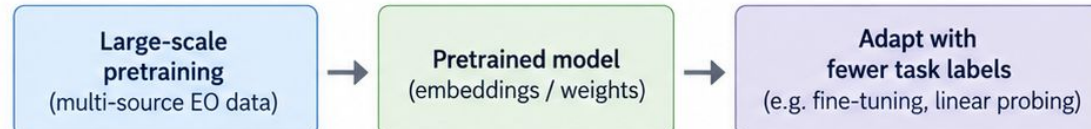


Train from scratch



Requires many task labels and model training for each task.

Foundation model approach



Leverages pretrained representations and typically needs fewer task labels.



Useful when pretrained representations transfer to the target sensor, region and task; independent validation is still required.

Pretraining aims to learn transferable EO representations that can later be adapted to specific tasks with fewer labels.

Why use foundation models, and what remains difficult?

Potential advantages

- Better use of large EO archives.
- Less reliance on large labelled datasets for every new task.
- Potential for transfer across tasks, regions or sensors.

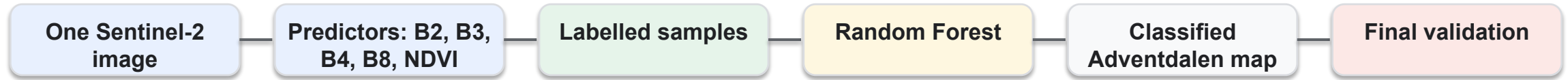
Important cautions

- Transfer is not automatic and may fail for a new region or sensor.
- Models can be difficult to interpret.
- Careful validation is still required.

State of the art — in one sentence

- Current GeoAI is moving towards EO-native pretraining, multimodal inputs and reusable embeddings, but research practice still depends on clear tasks, good reference data and rigorous evaluation.

How today's practical fits into the bigger picture



Why this design is useful for teaching

- It shows the complete workflow clearly.
- The map can be interpreted directly.
- The same logic scales to more advanced methods later.

Key points and selected references

Key points

- Classification turns EO measurements into a thematic map.
- Predictors, labels and validation are as important as model choice.
- Random Forest is a clear and robust baseline for the practical.
- Deep learning learns richer spatial representations from the imagery.
- Foundation models aim to learn reusable EO representations from large archives.
- State-of-the-art methods still require careful evaluation and ecological interpretation.

Selected references

- Belgiu, M. & Drăguț, L. (2016). Random Forest in remote sensing. ISPRS JPRS.
- Zhu, X. X. et al. (2017). Deep learning in remote sensing. IEEE GRSM.
- Xie, E. et al. (2021). SegFormer: efficient semantic segmentation.
- Recent EO foundation-model literature: Prithvi-EO-2.0 and TerraMind as examples of current directions.